

CHERLAK, Russian Federation

WMO: 287990

Lat: 54.167N

Lon: 74.800E

Elev: 115

StdP: 99.96

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.4	-30.7	-36.1	0.1	-33.3	-33.6	0.2	-30.7	7.5	-9.6	6.6	-8.9	1.4	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.0	19.2	29.2	18.8	27.4	18.0	20.8	27.8	19.9	26.5	19.1	25.5	2.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	23.3	17.5	12.7	22.6	16.7	12.1	22.1	60.6	28.1	57.6	26.6	55.0	25.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.1	5.3	DB	-35.6	34.0	3.5	1.5	-38.2	35.1	-40.2	36.0	-42.2	36.8	-44.7	37.9	(4)
(5)			WB	-35.8	22.5	3.4	1.7	-38.3	23.7	-40.3	24.7	-42.2	25.6	-44.7	26.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-18.2	-15.8	-7.3	4.7	13.2	18.5	19.9	17.5	11.4	4.1	-6.5	-14.5	(6)	
	DBStd	14.77	7.60	6.92	6.18	6.10	5.23	4.34	3.69	3.97	4.54	4.85	7.43	7.69	(7)	
	HDD10.0	3822	873	721	537	179	28	2	0	1	36	191	496	758	(8)	
	HDD18.3	6027	1132	954	796	409	174	51	28	65	213	442	746	1017	(9)	
	CDD10.0	1027	0	0	0	20	126	256	306	233	78	8	0	0	(10)	
	CDD18.3	188	0	0	0	0	14	55	75	39	5	0	0	0	(11)	
	CDH23.3	1847	0	0	0	6	193	544	674	364	66	0	0	0	(12)	
	CDH26.7	545	0	0	0	1	47	170	208	107	12	0	0	0	(13)	
(14) Wind	WSAvg	2.4	2.4	2.5	2.7	2.7	2.5	2.0	1.9	1.9	2.1	2.4	2.7	2.6	(14)	
(15) Precipitation	PrecAvg	365	20	17	19	19	24	43	64	48	25	31	30	25	(15)	
	PrecMax	537	38	27	48	54	82	89	144	130	70	73	73	54	(16)	
	PrecMin	225	2	2	4	2	2	6	9	6	2	4	8	7	(17)	
	PrecStd	76	10	7	13	13	16	21	40	28	15	18	16	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	0.9	7.2	23.6	30.4	32.5	33.7	32.6	28.2	20.1	7.9	1.1	(19)	
		MCWB	0.0	0.2	3.8	13.7	17.2	18.6	20.0	19.0	17.0	11.8	6.1	0.4	(20)	
	2%	DB	-2.0	-1.0	4.0	20.3	27.7	30.5	31.0	29.5	25.1	17.0	5.8	0.0	(21)	
		MCWB	-2.5	-1.6	2.0	11.8	15.8	18.8	19.7	19.3	15.2	10.6	4.3	-0.4	(22)	
	5%	DB	-4.1	-3.0	2.2	17.7	25.4	28.6	29.0	27.1	22.0	14.1	4.1	-1.7	(23)	
		MCWB	-4.6	-3.5	0.8	10.4	14.7	18.5	19.4	18.1	14.0	8.9	2.7	-2.2	(24)	
	10%	DB	-6.4	-5.2	1.0	14.7	22.8	26.5	27.1	25.0	19.3	11.7	2.1	-4.0	(25)	
		MCWB	-6.8	-5.8	0.0	8.9	13.5	17.6	19.0	17.4	12.8	7.8	1.0	-4.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.3	4.2	14.6	18.3	21.2	22.0	21.2	18.8	12.8	6.8	0.6	(27)	
		MCDB	0.8	0.7	6.7	22.4	27.9	29.7	28.0	28.0	27.4	18.3	7.8	0.9	(28)	
	2%	WB	-2.5	-1.6	2.2	12.4	16.8	20.0	21.1	20.1	16.1	11.3	4.4	-0.4	(29)	
		MCDB	-2.1	-1.1	3.6	18.6	26.1	27.3	27.7	26.9	22.4	15.9	5.5	0.0	(30)	
	5%	WB	-4.6	-3.5	1.1	10.7	15.5	19.1	20.3	19.2	14.6	9.6	2.8	-2.3	(31)	
		MCDB	-4.1	-3.0	2.1	16.9	23.6	26.2	26.7	25.3	21.0	13.7	4.0	-1.7	(32)	
	10%	WB	-6.9	-5.8	0.0	9.2	14.2	18.1	19.5	18.3	13.1	8.0	1.0	-4.6	(33)	
		MCDB	-6.5	-5.3	1.0	14.5	21.3	25.2	25.7	23.8	18.5	11.4	2.0	-4.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.6	8.0	9.7	11.8	10.8	10.0	10.3	10.2	7.9	6.2	7.2	(35)	
		MCDBR	8.4	7.1	7.0	14.4	15.1	13.7	12.7	13.5	14.2	12.8	6.5	7.4	(36)	
	5% WB	MCWBR	8.4	7.0	5.7	8.1	6.7	5.6	4.7	5.8	6.9	7.4	5.4	7.5	(37)	
		MCWBR	8.4	7.1	6.8	13.4	13.7	11.7	10.8	11.4	13.0	11.6	6.1	7.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.319	0.378	0.405	0.404	0.412	0.435	0.428	0.369	0.352	0.328	0.292	(40)	
		taud	2.045	2.013	1.932	2.148	2.219	2.280	2.211	2.196	2.320	2.250	2.106	2.009	(41)	
	Ebn at Noon		624	737	769	816	841	837	807	786	793	706	567	519	(42)	
		Edh at Noon	72	105	143	133	131	125	131	126	98	83	68	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.07	2.14	3.68	4.66	5.75	6.13	5.59	4.61	3.36	1.74	0.83	0.69	(44)	
	RadStd		0.08	0.09	0.29	0.37	0.51	0.38	0.39	0.35	0.29	0.23	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page